

FREDERIKSHAVN, Denmark

WMO: 060430

Lat: 57.4361N

Lon: 10.5489E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 91-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.8	-5.3	-12.1	1.3	-3.4	-10.5	1.5	-2.6	16.4	4.2	15.0	4.5	6.0	40	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.5	22.4	17.9	21.3	17.2	20.1	16.5	19.2	21.1	18.2	20.3	17.2	19.4	3.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.2	20.4	17.3	12.4	19.5	16.2	11.5	18.6	54.5	21.2	51.4	20.4	48.3	19.6	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.1	12.5	11.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB													
			WB	-9.0	20.0	2.5	1.5	-10.8	21.0	-12.3	21.9	-13.7	22.7	-15.5	23.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	1.1	0.4	2.3	5.9	10.4	13.0	15.5	16.1	12.8	8.3	4.1	1.7
	DBStd	6.22	3.16	3.39	2.48	2.89	2.94	2.47	2.55	2.67	2.24	2.81	2.75	3.28
	HDD10.0	1447	276	268	237	128	31	2	0	0	3	67	177	256
	HDD18.3	3907	534	501	496	373	247	161	95	81	167	311	427	514
	CDD10.0	599	0	0	0	6	43	92	169	189	86	14	0	0
	CDD18.3	19	0	0	0	0	0	1	6	12	0	0	0	0
	CDH23.3	12	0	0	0	0	0	1	3	8	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.8	6.7	6.5	6.2	5.0	5.2	5.8	5.1	5.0	5.5	6.1	6.3	6.2
Precipitation	PrecAvg	720	60	42	38	38	47	71	63	75	74	79	62	60
	PrecMax	976	113	81	96	93	97	149	153	136	266	146	111	119
	PrecMin	487	9	10	4	10	5	1	7	6	36	11	17	16
	PrecStd	116	28	22	22	24	22	35	38	34	46	36	26	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	8.9	10.0	16.8	19.5	21.6	23.7	24.4	20.5	15.0	11.3	9.2
		MCWB	7.1	6.8	5.6	11.7	13.4	16.4	18.5	17.5	16.4	12.8	9.9	8.2
	2%	DB	7.1	7.1	8.0	13.7	17.7	20.1	22.3	22.7	18.7	13.6	10.0	8.1
		MCWB	6.1	5.9	5.6	9.8	12.4	15.1	17.9	18.2	15.2	11.9	8.7	7.1
	5%	DB	6.2	6.0	6.8	11.6	16.4	18.6	21.1	21.6	17.2	12.8	9.0	7.4
		MCWB	5.2	4.9	4.9	8.5	12.2	14.3	17.4	17.5	14.3	11.5	7.8	6.5
	10%	DB	5.3	5.2	5.7	9.9	15.0	17.2	19.9	20.6	15.9	12.1	8.1	6.4
		MCWB	4.3	4.0	4.2	7.4	11.6	13.4	16.5	17.0	13.7	10.8	6.9	5.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	7.7	7.3	12.4	15.0	17.8	20.7	20.2	17.4	13.5	10.5	8.5
		MCDB	8.4	8.5	8.2	15.9	17.5	20.4	22.5	22.3	19.3	14.4	11.2	8.9
	2%	WB	6.3	6.2	6.3	10.5	14.1	16.2	19.5	19.2	16.1	12.5	9.0	7.3
		MCDB	6.8	6.8	7.3	12.6	16.3	18.8	21.2	21.2	18.0	13.2	9.7	7.8
	5%	WB	5.4	5.1	5.5	9.0	13.1	15.1	18.2	18.5	14.9	11.9	8.1	6.6
		MCDB	6.1	5.8	6.4	10.9	15.8	17.5	20.2	20.5	16.4	12.7	8.8	7.3
	10%	WB	4.5	4.2	4.6	7.9	11.9	14.1	16.9	17.7	14.1	11.1	7.1	5.7
		MCDB	5.2	5.0	5.6	9.5	14.3	16.5	19.0	19.9	15.4	12.0	8.0	6.4
Mean Daily Temperature Range	5% DB	MDBR	3.1	3.4	3.8	4.5	5.4	5.9	6.0	5.5	4.1	3.6	2.9	3.1
		MCDBR	3.9	4.3	5.8	6.6	7.1	7.3	7.1	6.1	5.9	4.1	4.1	3.8
		MCWBR	3.9	3.6	4.1	3.4	4.3	4.2	3.9	3.9	3.5	3.2	3.9	3.7
	5% WB	MCDBR	3.7	4.1	5.0	6.1	6.2	6.4	5.7	5.4	4.8	3.8	3.8	3.9
		MCWBR	3.9	4.0	4.1	3.6	4.3	4.7	4.2	4.1	3.3	3.3	3.8	3.8
Clear-Sky Solar Irradiance	taub	0.306	0.332	0.349	0.387	0.379	0.369	0.390	0.391	0.362	0.347	0.323	0.291	
	taud	2.245	2.305	2.344	2.294	2.386	2.440	2.401	2.401	2.467	2.418	2.298	2.205	
	Ebn at Noon	548	705	801	823	857	871	841	813	785	685	524	450	
	Edh at Noon	51	73	91	112	109	104	106	100	81	65	48	39	
All-Sky Solar Radiation	RadAvg	0.37	0.93	2.38	4.06	5.42	6.01	5.51	4.30	2.92	1.35	0.49	0.26	
	RadStd	0.07	0.17	0.23	0.46	0.43	0.43	0.51	0.28	0.28	0.18	0.08	0.03	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-120	-165	N/A	N/A

Nomenclature: See separate page